EE511 Protection of Power Systems II

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- **Slides 10-18 are from Chapter 7 of "Digital Signal Processing in Power System Protection and Control", Springer, 2011, by Waldemar Rebizant, Janusz Szafran, Andrzej Wiszniewski**

Detection of the Unsaturated Fragment of the Waveshape

An exemplary detailed waveshape of the CT saturated secondary current is shown in Fig. 7.3, that can be used as illustration and basis for the developed saturation detection methods.

One may see that the CT saturates between the samples $(n-1)$ and (n). It means that the core flux reached the saturation level. The saturation ends between the samples $(m-1)$ and (m). Therefore, the transformation is correct, which means that the secondary current is proportional to the primary current ($i_2 = N_i i_1$), until the sample (n), and after the sample (m).

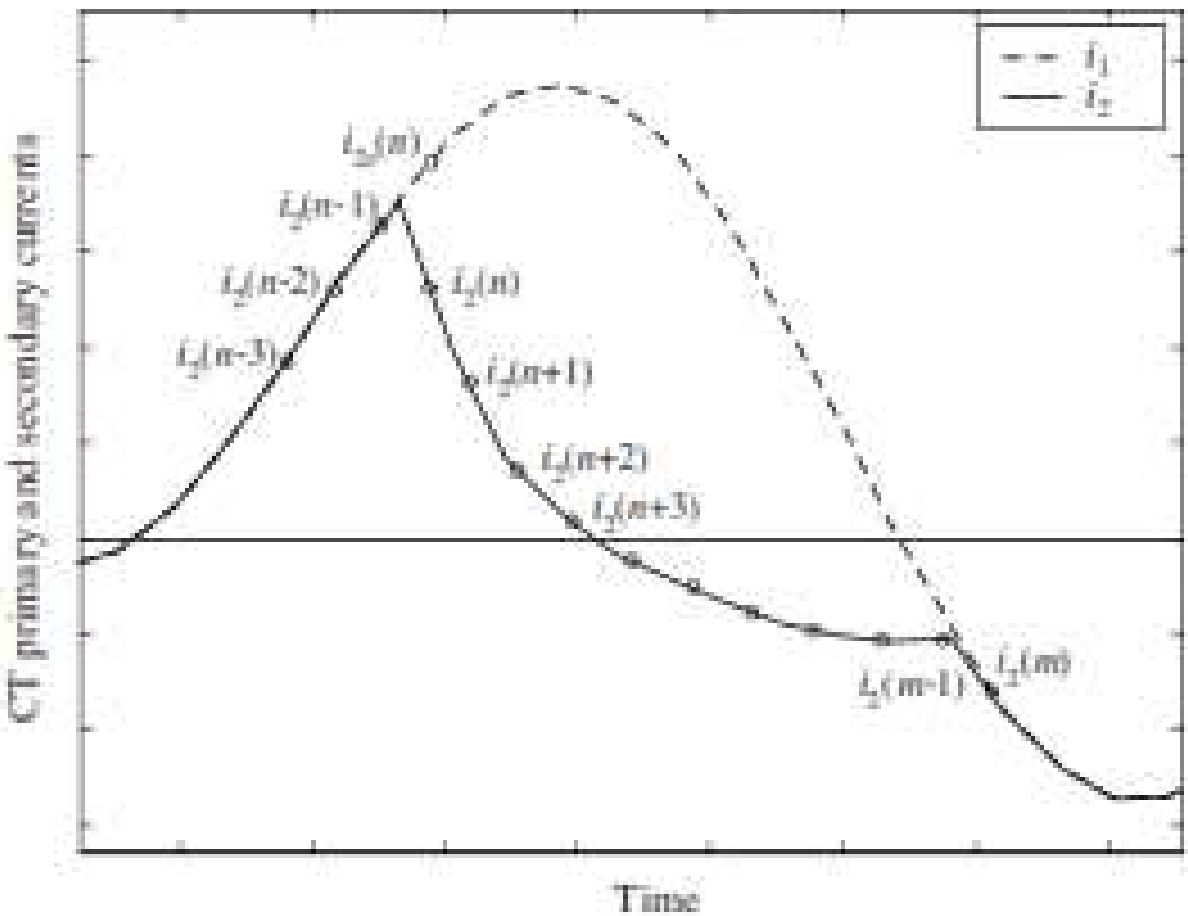

Fig. 7.3 Transient waveform of the saturated CT secondary current

Determination of the sample (n) is based on the comparison of the measured secondary current $i_2(n)$ and the estimated secondary current $i_{2e}(n)$:

$$|i_{2e}(n) - i_2(n)| \geq \Delta |i_{2e}(n)| \tag{7.4}$$

and

$$|i_{2e}(n)| - |i_2(n)| \geq 0 \tag{7.5}$$

If the difference between the two exceeds a part Δ of the estimated level, it means that there is a sudden change of the waveshape and saturation happened between the samples.

If the condition (7.4) is satisfied, but

$$|i_{2e}(n)| - |i_2(n)| \leq 0 \tag{7.6}$$

like between the samples (m) and $(m-1)$ in Fig. 7.3, it marks the end of saturation.

Estimation of $i_{2e}(n)$ may be done with the assumption that the derivatives of the secondary current between the samples $(n-1)$ and $(n-2)$ are the same, as between the samples (n) and $(n-1)$. The simplest way is to assume that the first order derivatives are equal:

$$\frac{i_{2e}(n) - i_2(n-1)}{T_S} \approx \frac{i_2(n-1) - i_2(n-2)}{T_S}, \tag{7.7}$$

which leads to the formula:

$$i_{2e}(n) \approx 2i_2(n-1) - i_2(n-2). \tag{7.8}$$

And now one may apply the condition (7.4) to find out whether there was a rapid change in the waveshape.

Second and third order derivatives can be calculated with the well known formulae:

$$i''(n) \approx \frac{i'(n) - i'(n-1)}{T_S}, \tag{7.9}$$

$$i'''(n) \approx \frac{i''(n) - i''(n-1)}{T_S}. \tag{7.10}$$

If one assumes that the second order derivative is the same in nth sampling period, then the value of $i_{2e}(n)$ may be estimated by the formula:

$$i_{2e}(n) \approx 3i_2(n-1) - 3i_2(n-2) + i_2(n-3), \tag{7.11}$$

which is far more accurate than (7.8). However, if one wants to increase accuracy even more, one may assume that the third derivative is to be the same between the samples, what gives the condition:

$$i_{2e}(n) \approx 4i_2(n-1) - 6i_2(n-2) + 4i_2(n-3) - i_2(n-4), \tag{7.12}$$

The same procedure may be repeated to estimate the sample $i_{2e}(m)$ what marks the end of saturation.

Correction of the Secondary Current

Equivalent circuit of the CT reduced to the secondary side is presented in Fig. 7.4 and the magnetizing characteristic of the core in Fig. 7.5.

Estimation of the true values of the secondary current by means of the formula (7.12) is sufficiently accurate while calculating a maximum of two samples of the current after saturation. Therefore, one may assume that:

$$i_{2e}(n) \approx i_1(n),$$

$$i_{2e}(n+1) \approx i_1(n+1).$$

Thus, the samples of the magnetizing current become:

$$i_m(n) \approx i_{2e}(n) - i_2(n), \tag{7.20}$$

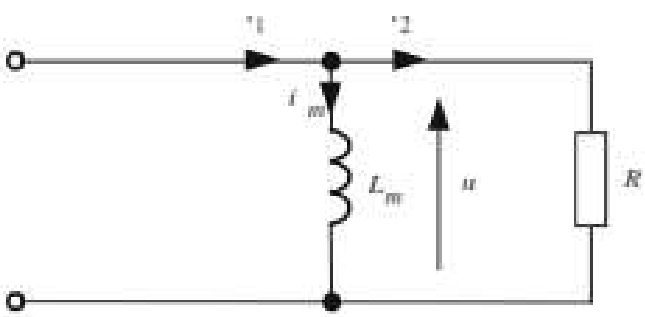

Fig. 7.4 Simplified equivalent circuit of the saturated CT

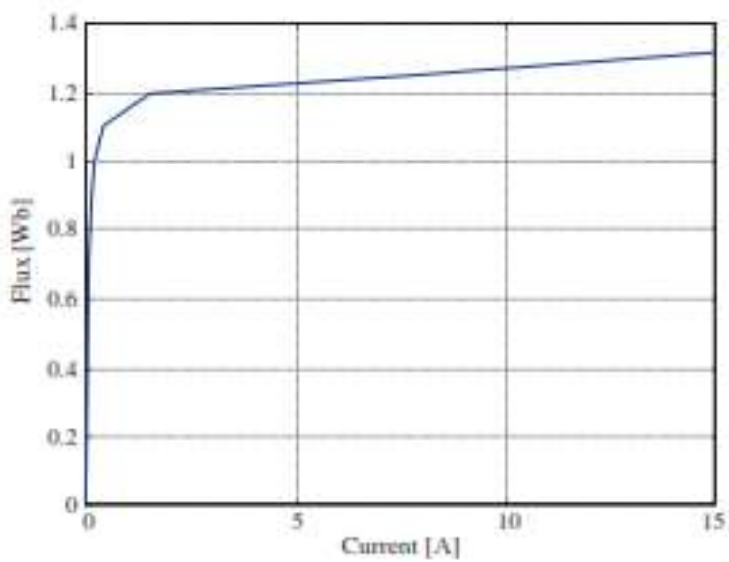

Fig. 7.5 Magnetizing characteristic of the CT core

$$i_m(n+1) \approx i_{2r}(n+1) - i_2(n+1). \tag{7.21}$$

Then the change of the magnetizing current between the samples (n) and $(n+1)$ becomes:

$$\Delta i_m(n+1) \approx i_m(n+1) - i_m(n). \tag{7.22}$$

The mean value of the CT secondary voltage between the samples (n) and $(n+1)$ equals:

$$u(n+1) \approx R_2 \frac{i_2(n+1) + i_2(n)}{2} + L_2 \frac{i_2(n+1) - i_2(n)}{T_S}. \tag{7.23}$$

Therefore, the increase of the flux linkage Ψ in one sampling period becomes:

$$\Delta\psi(n+1) \approx T_S u(n+1). \tag{7.24}$$

The value of the magnetizing inductance between the samples is given by the formula:

$$L_m(n+1) \approx \frac{\Delta\psi(n+1)}{\Delta i_m(n+1)}. \tag{7.25}$$

Now, assuming that between the samples $(n+1)$ and $(n+2)$ the magnetizing inductance L_m has the same value, as during the previous sample, one may write:

$$\Delta i_m(n+2) \approx \frac{\Delta\psi(n+2)}{L_m(n+1)} \approx \Delta i_m(n+1)\frac{\Delta\psi(n+2)}{\Delta\psi(n+1)}, \tag{7.26}$$

where $\Delta\psi(n+2) \approx T_S u(n+2)$.

Further, for the next time instant one obtains:

$$u(n+2) \approx R_2 \frac{i_2(n+2) + i_2(n+1)}{2} + L_2 \frac{i_2(n+2) - i_2(n+1)}{T_S}, \tag{7.27}$$

therefore:

$$i_m(n+2) \approx i_m(n+1) + \Delta i_m(n+2) \tag{7.28}$$

and in consequence:

$$i_{2r}(n+2) \approx i_2(n+2) + i_m(n+2). \tag{7.29}$$

This algorithm may be repeated, bearing in mind that:

$$L_m(n+2) \approx \frac{\Delta\psi(n+2)}{\Delta i_m(n+2)}, \tag{7.30}$$

$$\Delta i_m(n+3) \approx \Delta i_m(n+2) \frac{\Delta\psi(n+3)}{\Delta(n+2)}, \tag{7.31}$$

and consequently:

$$i_{2r}(n+3) \approx i_2(n+3) + i_m(n+3). \tag{7.32}$$

The process of calculation of the estimated values of the secondary current samples continues, i.e.,

$$\Delta i_m(n+k) \approx \Delta i_m(n+k-1)\frac{\Delta\psi(n+k)}{\Delta\psi(n+k-1)}, \tag{7.33}$$

$$i_{2e}(n+k) \approx i_2(n+k) + i_m(n+k), \tag{7.34}$$

until the end of the saturated fraction of the period